

Structural Equation Modeling workshop

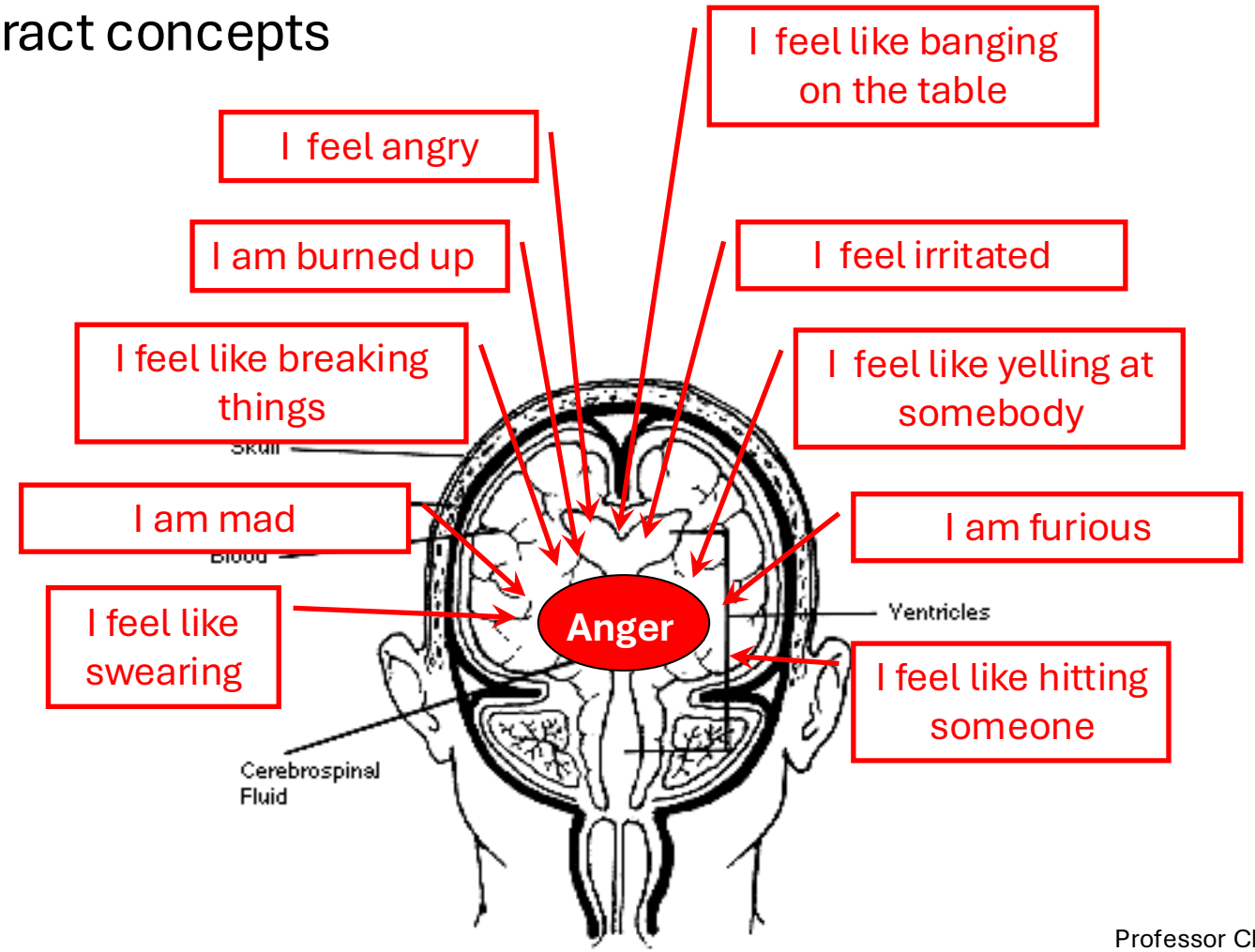
Mathew Marques
School of Psychology and Public Health
La Trobe University
30 October 2026

Model-based testing: Factor Analysis

Part 1 of 4

WHAT IS FACTOR ANALYSIS ?

- Well… what is psychological measurement ?
  - Psychological measurement is about trying to assess hypothetical abstract concepts



REMEMBER!

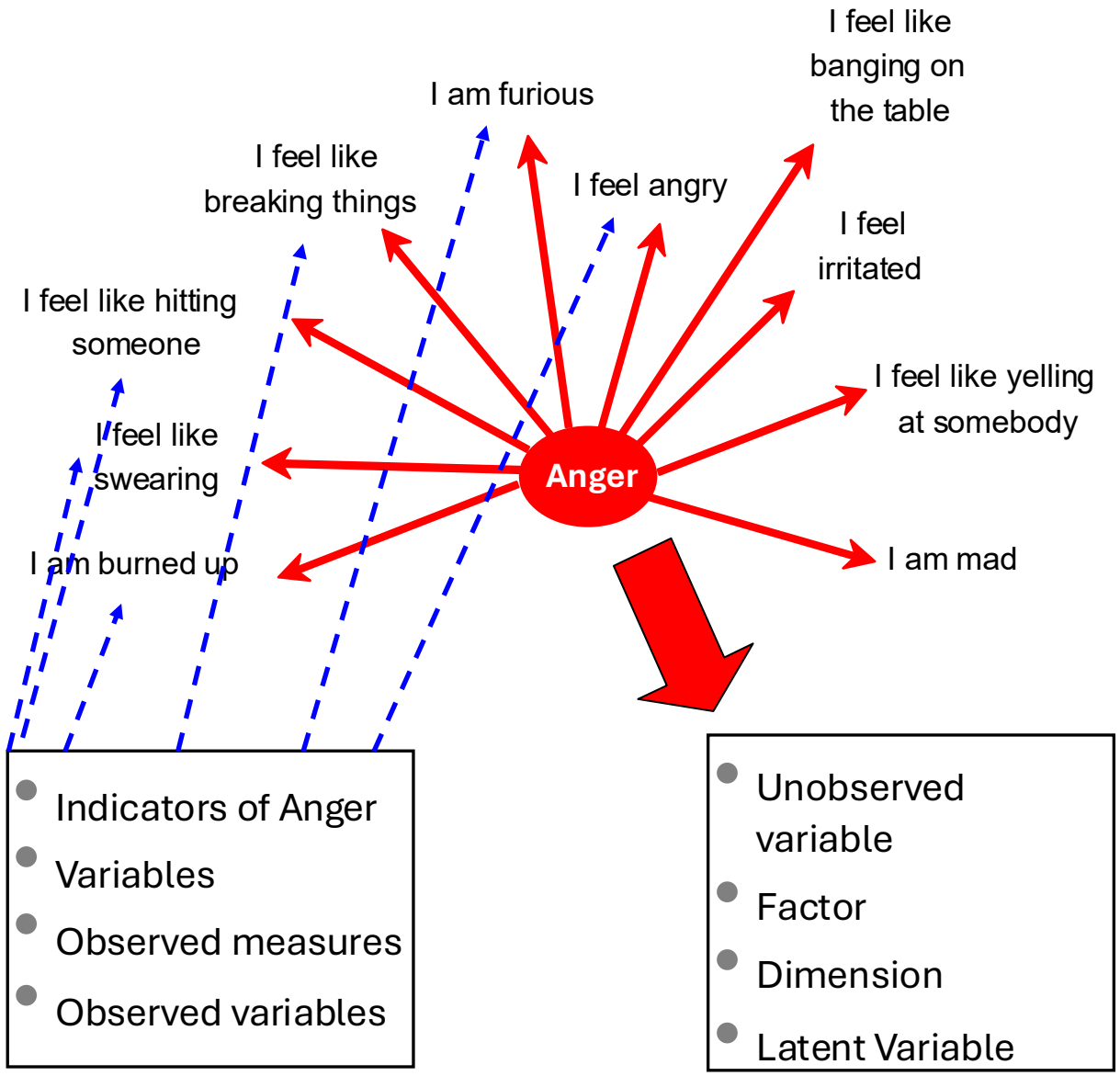
- Psychological constructs are abstract and hypothetical.
- We can never directly or physically measure them.
- Scores on psychological tests rarely provide absolute, ratio scale measures of psychological attributes.
- All we can ever do is compare our scores to other scores (i.e., with norms).

OR

- Examine the pattern of relationships between variables or observations to see if they fit with our theories.

WHAT IS FACTOR ANALYSIS ?

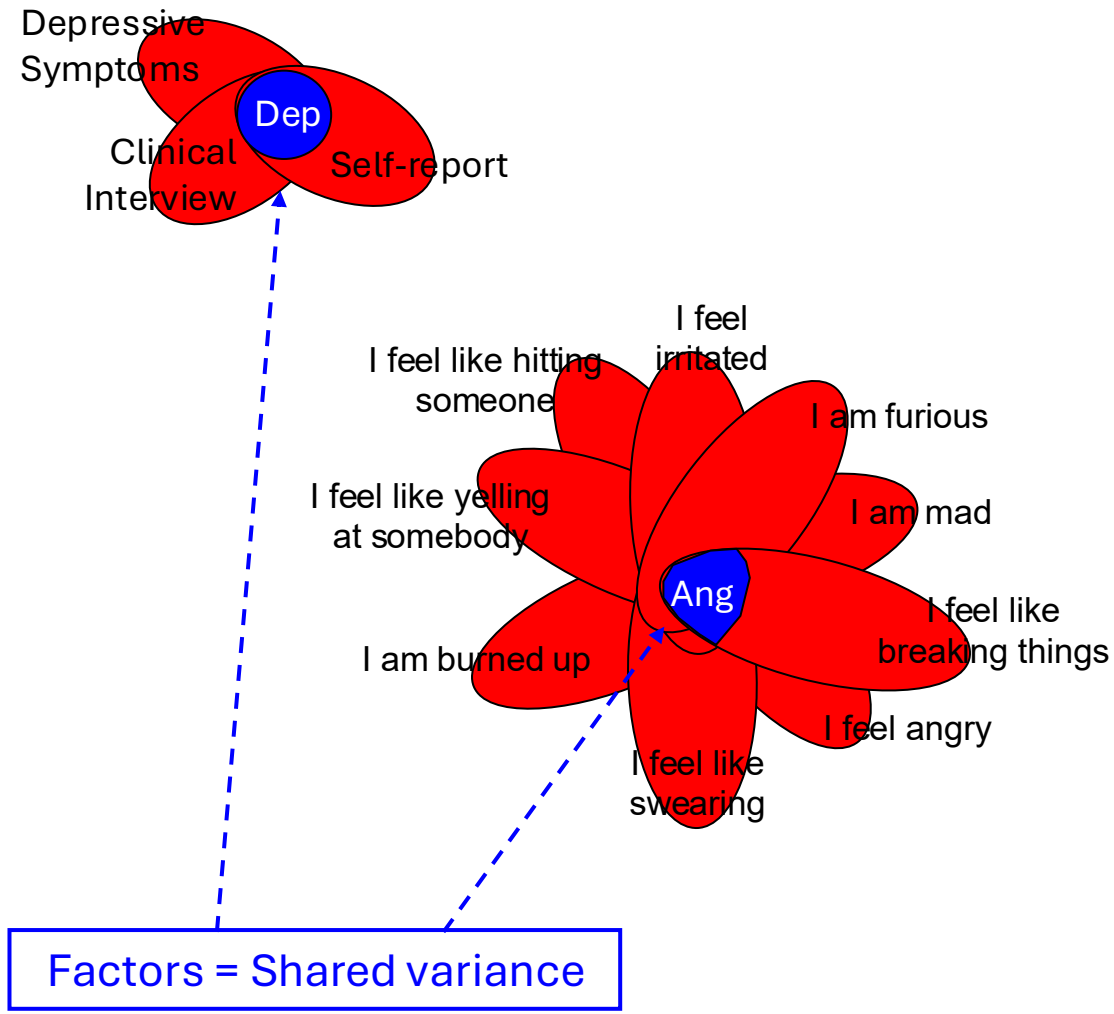
A statistical technique designed to reduce a number of common characteristics into one or more dimensions or factors.



AN OVERVIEW OF FACTOR ANALYSIS

- Factor analysis is a statistical technique to examine the underlying patterns of correlations in a set of variables.
- The method is used for a variety of purposes, including to better understand the structure of your data as well as testing some specific prior expectations (model) about your data.
- Two types of factor analysis are:
  - Exploratory factor analysis (EFA) – to explore the data structure (how many factors; their relations and effects on each variable).
  - Confirmatory factor analysis (CFA) – to test a theory on data structure.
- Two types of EFA are:
  - Principal Components Analysis (PCA) – data-driven; for data reduction.
  - Common Factor Analysis – theory-driven; construct-index relationships.

CONCEPTUALLY HOW DOES FACTOR ANALYSIS WORK?



STATISTICALLY HOW DOES FACTOR ANALYSIS WORK?

- The pattern of shared variance (i.e., relationships) among the variables must correspond to the hypothesised pattern of the constructs in the ‘real world’.

- What are some pragmatic issues?
  - See Tabachnick & Fidell (Ch 13.3.2.1 & 14.3.2.1)

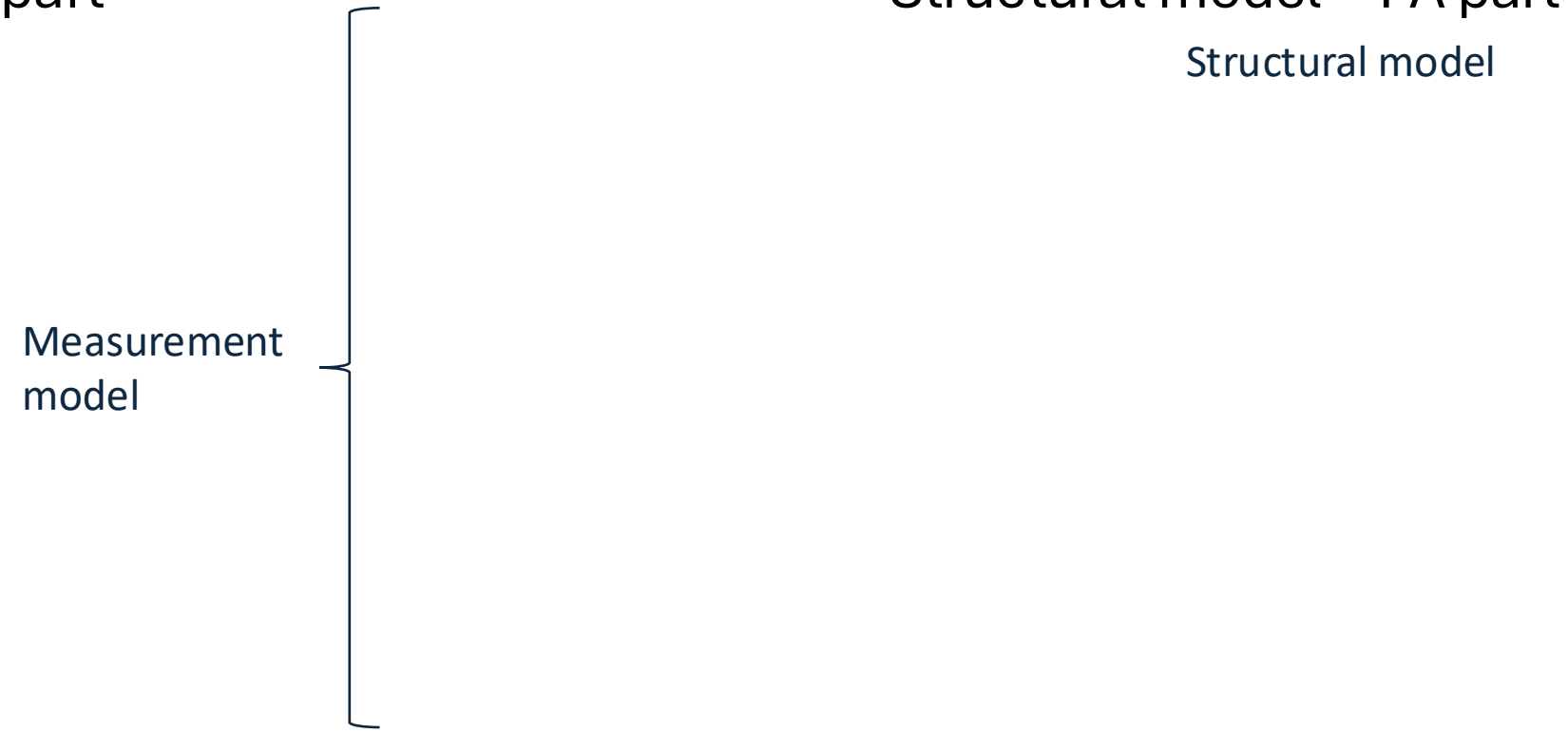
SO THEN, WHAT IS CONFIRMATORY FACTOR ANALYSIS ?

- Factor analysis: “The primary goal is to explain the covariances or correlations between many observed variables by means of relatively few underlying latent variables. In this sense it is a data reduction technique.” (Bollen, 1989)

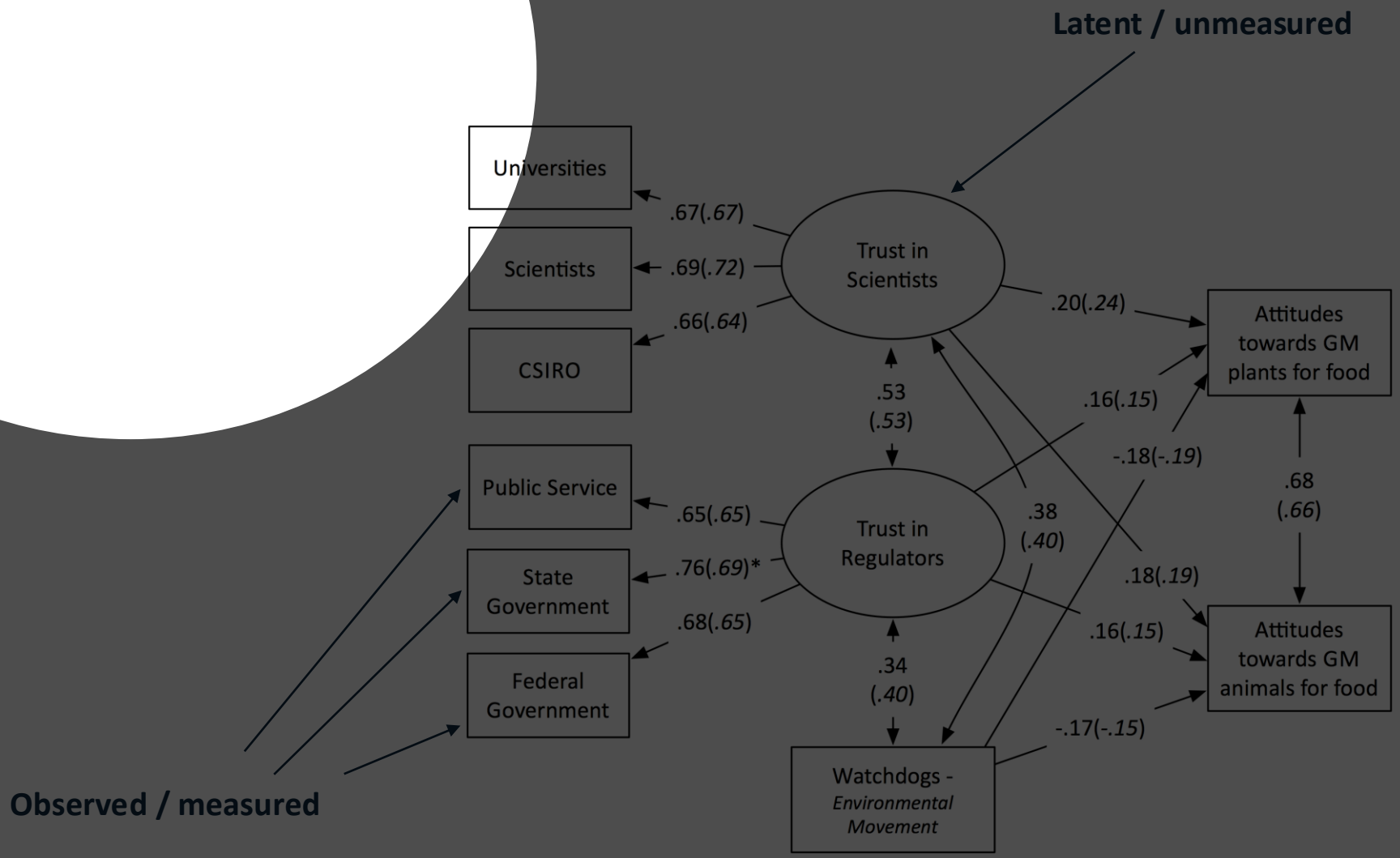
- CFA: a special form of factor analysis used to test how well the measured variables represent factors or constructs.

STRUCTURAL EQUATION MODEL

- A combination of CFA and PA
- Measurement model = CFA part
- Structural model = PA part



STRUCTURAL EQUATION MODEL



Marques, M. D., Critchley, C. R., & Walshe, J. (2015). Attitudes to genetically modified food over time: How trust in organizations and the media cycle predict support. *Public Understanding of Science*, *24*(5), 601-618.

CONFIRMATORY FACTOR ANALYSIS

In CFA, analysts specify which factor each variable should load based on the theory and evaluate the overall fit of the model.

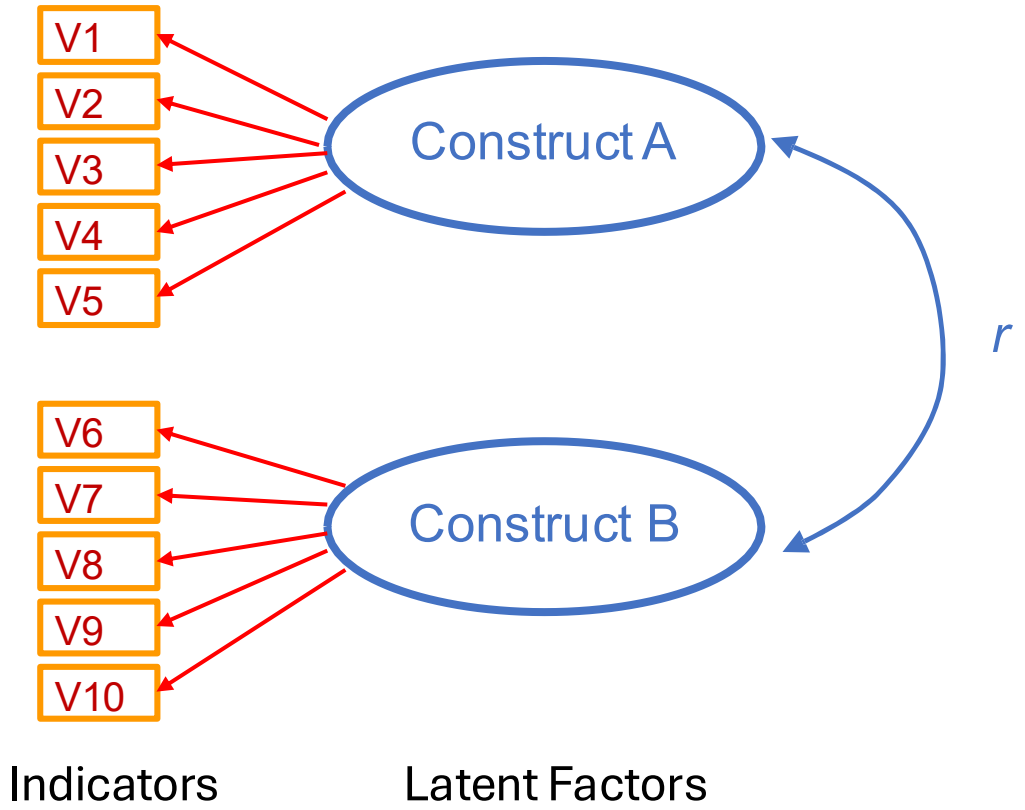
- In EFA, which variable should load on which factor is not tested rigorously. The overall fit of the model is not (usually) tested.

- CFA is considered a type of (or a part of) Structural Equation Modelling (SEM).

- CFA (measurement model) and path analysis (structural model) form a complete Structural Equation Model (SEM).

EXAMPLE OF TWO-FACTOR CONFIRMATORY FACTOR ANALYSIS

In CFA, variables are indicators of latent factors (constructs).
Constructs cannot be measured directly.



FIT INDICES FOR CFA / SEM

- Fit indices establish whether the model is acceptable.
  - If the model is acceptable, only then you can interpret the model.
- But what is an acceptable model fit?
- Need to look at fit indices as chi-square value (χ^2) is often statistically significant (not a good fit!). Sensitive to Type I error.
- Using a range of fit indices, often derived from the chi-square value (χ^2) - which represents the differences between the observed covariance matrix and the predicted/model covariance matrix – can allow us to determine whether a model is acceptable.

FIT INDICES FOR CFA/SEM

- Chi Square test χ^2 : too liberal, but you want it non-significant.
- Indices that compare the target and null models
  - Comparative Fit Index (CFI): > .93 (Byrne, 1994)
  - Incremental Fit Index (IFI): > .90
  - Tucker Lewis index (TLI): > .90
- Discrepancy indices
  - Standardized Root Mean Square Residual (SRMR): <.08 (Hu & Bentler, 1999)
  - Root Mean Square Error of Approximation (RMSEA): <.05 with UL CI <.08 (Hu & Bentler, 1999)
- Fit indices overestimate the fit when the sample size is small (e.g., < 200). *RMSEA* and *CFI* seem to be less sensitive to sample size (Fan, Thompson, & Wang, 1999).

- Barrett, P. (2007). Structural equation modelling: Adjudging model fit. *Personality and Individual differences, 42*(5), 815-824. https://doi.org/10.1016/j.paid.2006.09.018
- Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: conventional criteria versus new alternatives. Structural Equation Modeling, *6*(1), 1–55. https://doi.org/10.1080/10705519909540118

THE FIT OF THE MODEL IN CFA

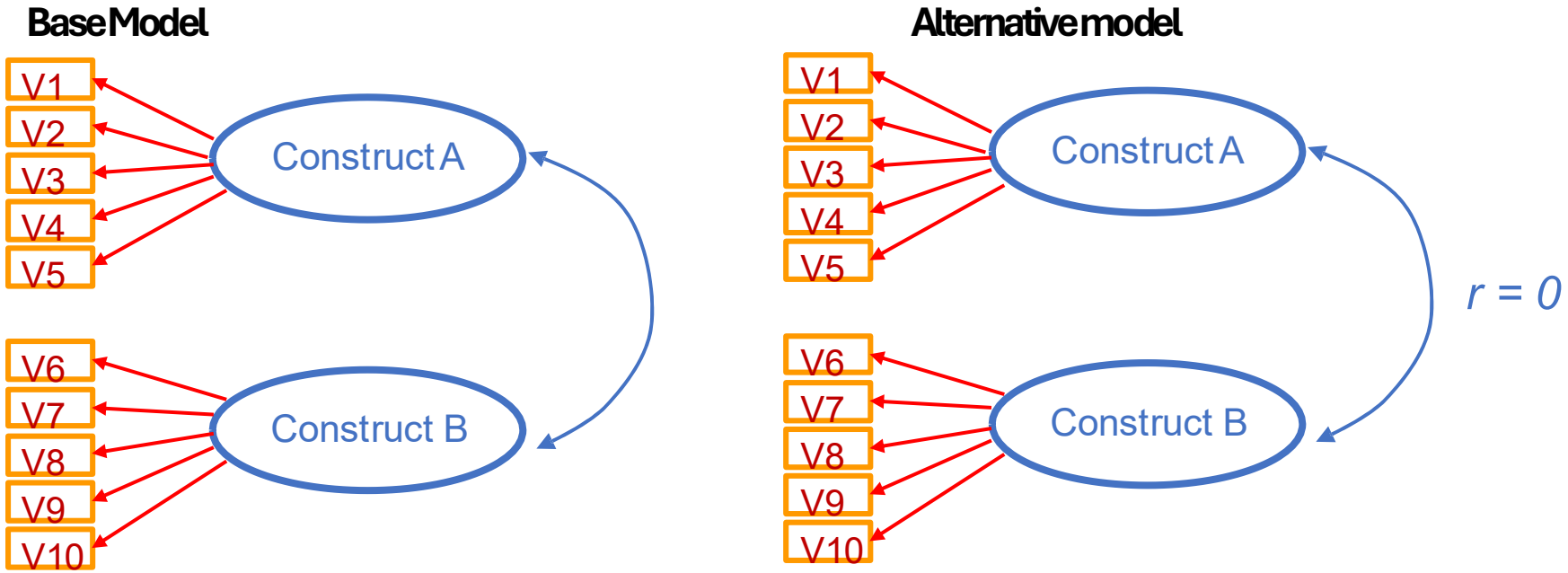
- The overall fit of the model is evaluated by using several fit indices (to compensate for limitations of each index):

Index	Recommended cut-off	
χ^2	Nonsignificant	
χ^2/ df	< 5.0 [a]	
RMSEA	< .08 [b, c]	
SRMR	< .08 [b, c]	
CFI	> .95 [b]	> .93 [d]
TLI	> .95 [b]	> .90 [d]

[a] Wheaton, Muthén, Alwin & Summers (1977)
[b] Putnick & Bornstein (2016)
[C] Hu & Bentler (1999)
[d] Byrne (1994)

THE FIT OF THE MODEL IN CFA

- Incremental fit indices (e.g., CFI, TLI) are comparing the base / specified model against a baseline or alternative model;
- “Is the alternative model significantly better/worse than the base model?”

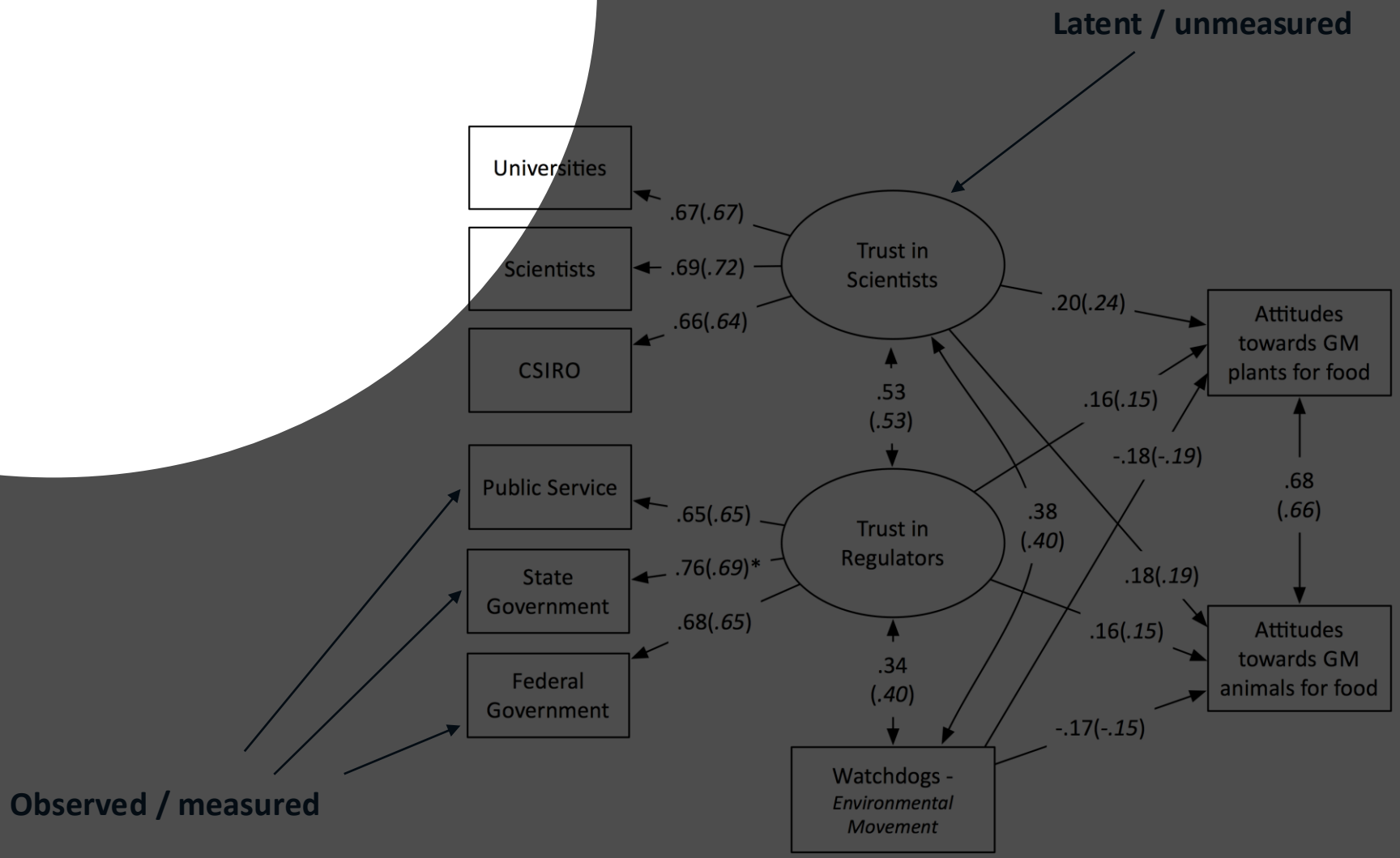


The significance of the predicted factor loading is also tested.

Model-based testing: Path Analysis

Part 2 of 4

Structural Equation Modelling (SEM)



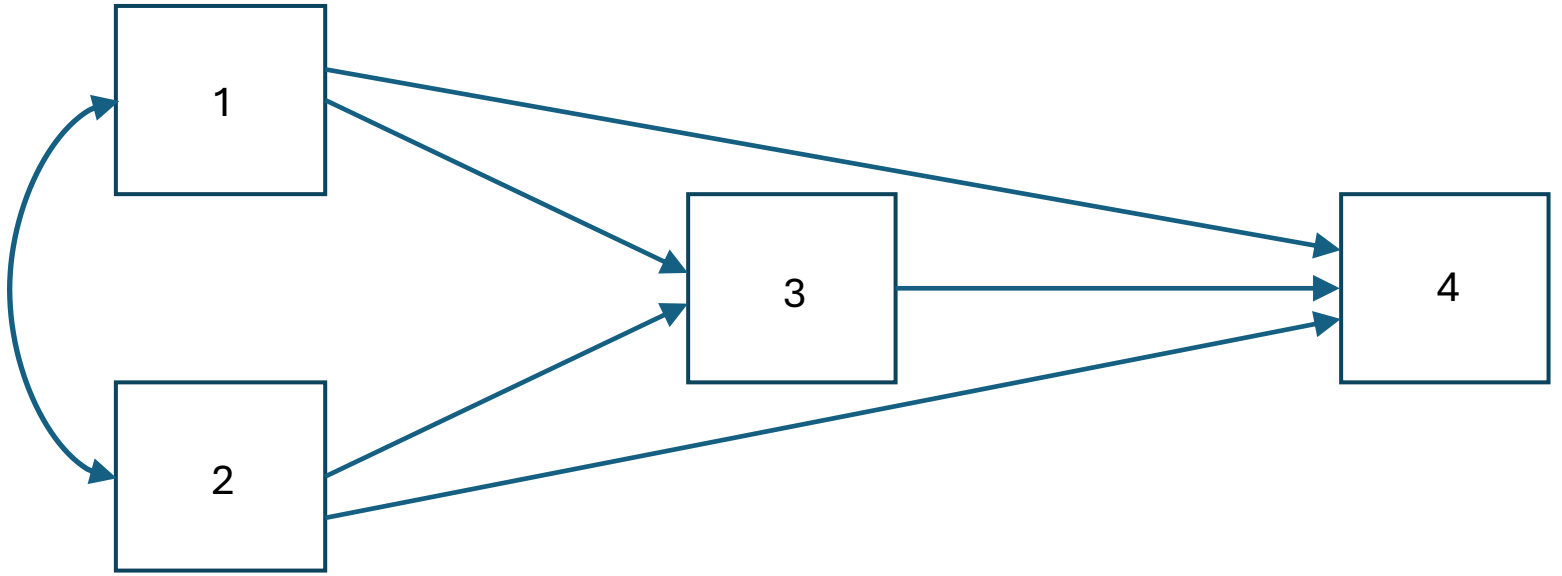
Marques, M. D., Critchley, C. R., & Walshe, J. (2015). Attitudes to genetically modified food over time: How trust in organizations and the media cycle predict support. *Public Understanding of Science*, *24*(5), 601-618.

PATH ANALYSIS – WHAT IS IT?



https://en.wikipedia.org/wiki/Sewall_Wright#/media/File:Sewall_Wright-en.jpg

- Developed by geneticist Sewall Wright (~1918).
- An extension of multiple regression focusing on causality.
- Special case of Structural Equation Modeling (SEM).
  - Only single indicators used for each variable in the causal model.
  - Path analysis is SEM with the structural model, but has no measurement model. Therefore, it assumes no measurement error.



ASSUMPTIONS FOR PATH ANALYSIS

Requires the usual assumptions of multiple regression.

- **Linearity**: Each predictor's relationship with the outcome is adequately represented by a straight-line effect.
- **Additivity**: Predictors' effects combine additively; interactions are absent unless explicitly included.
- **Interval-level data**: Variables are measured on scales where differences are meaningfully interpreted as equal. (In practice, ordinal variables may sometimes be modeled with suitable methods.)
- **Uncorrelated residuals**: Unmeasured influences on the outcome are not correlated with one another, unless correlated errors are explicitly modeled and justified.
- **No reverse or spurious causation**: For causal interpretations, the proposed direction of effects should be plausible, and associations should not be explained by reverse direction or omitted common causes. Path analysis alone cannot establish causality.
- **Low multicollinearity**: Predictors are not so highly correlated that their separate effects become difficult to estimate.

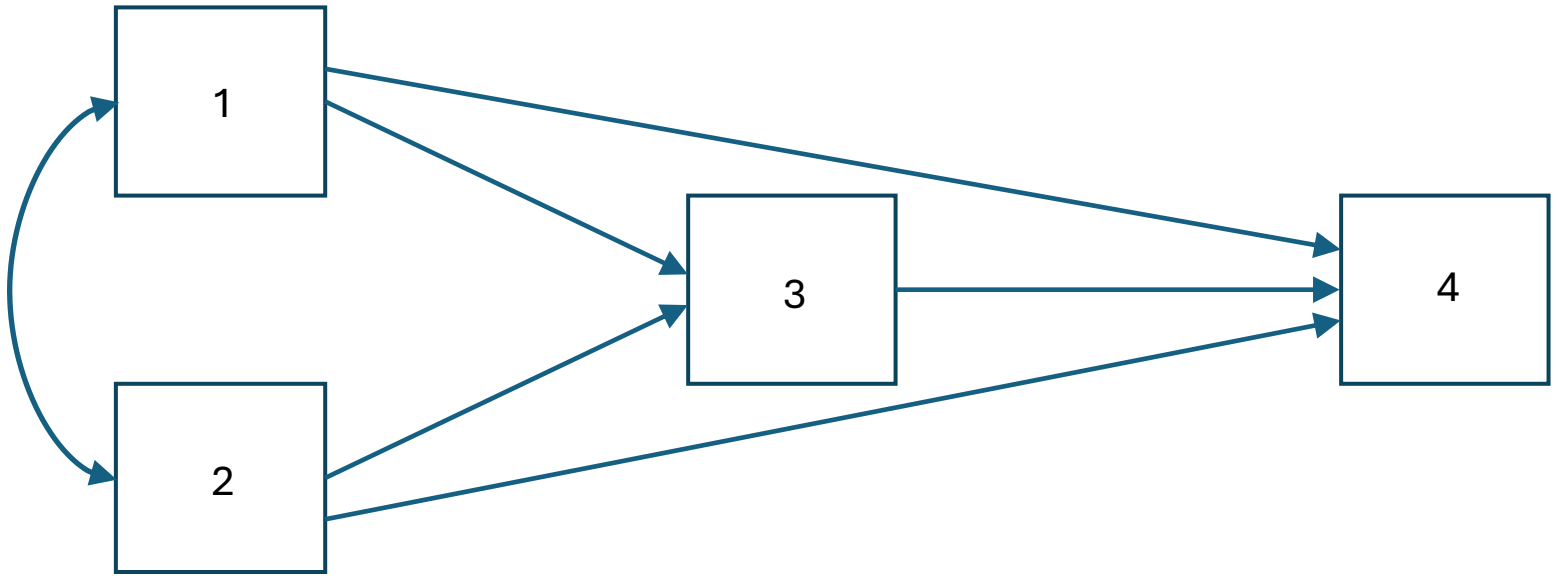
Additional assumptions of path analysis.

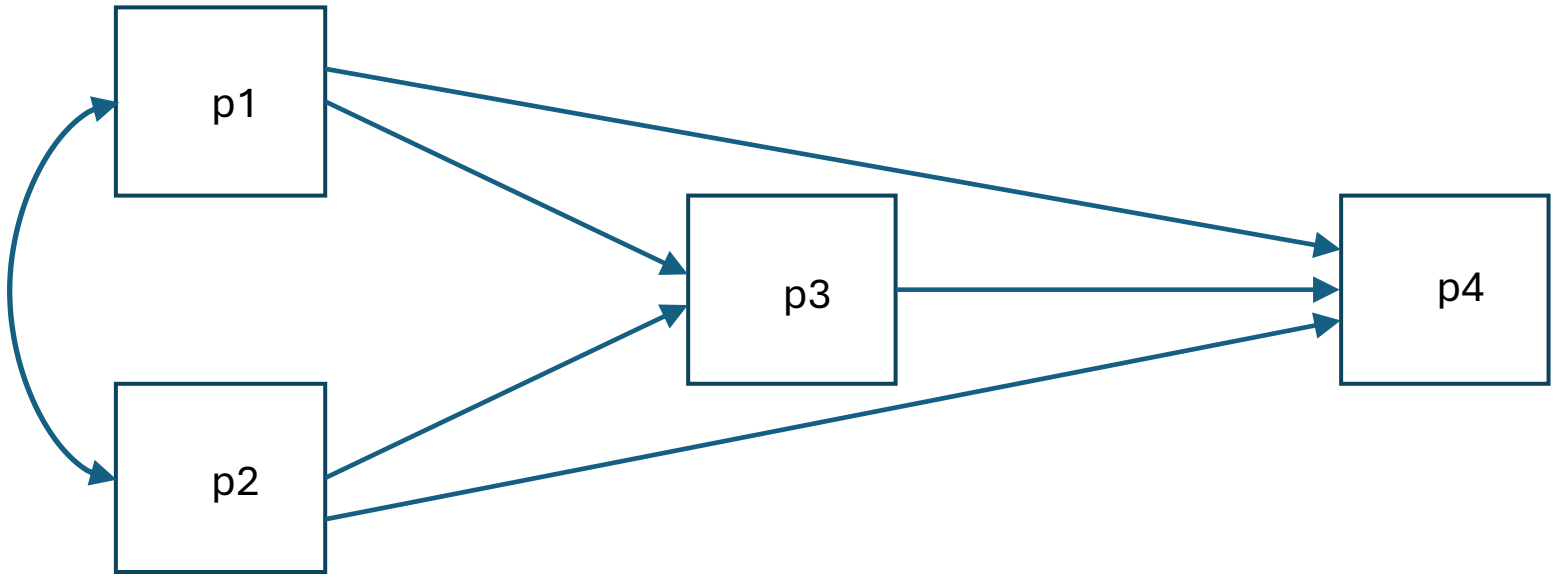
- **No under-identification or underdetermination**: The data provide enough information to estimate all model parameters uniquely. An under-identified model has no unique solution.
- **Proper specification**: The model's variables, paths, and error relationships reflect a defensible theory; omitted or incorrectly included paths can distort results.
- **Appropriate correlation input**: The input matrix must suit the variables and estimator—for example, covariance data for unstandardized estimates, or an appropriate correlation matrix for standardized analyses.
- **Adequate sample size**: There are enough observations for stable estimates and reliable tests; the needed number depends on the model and data.
- **Same sample**: The correlations or covariances used together should be based on the same set of cases, so they represent one consistent sample.

IT'S JUST LIKE HIERARCHICAL MULTIPLE REGRESSION

- Exogenous variable: variance assumed to be caused entirely by variables not in the causal model.

- Endogenous variable: variance assumed to be explained in part by other variables in the model.


1
2
3
4

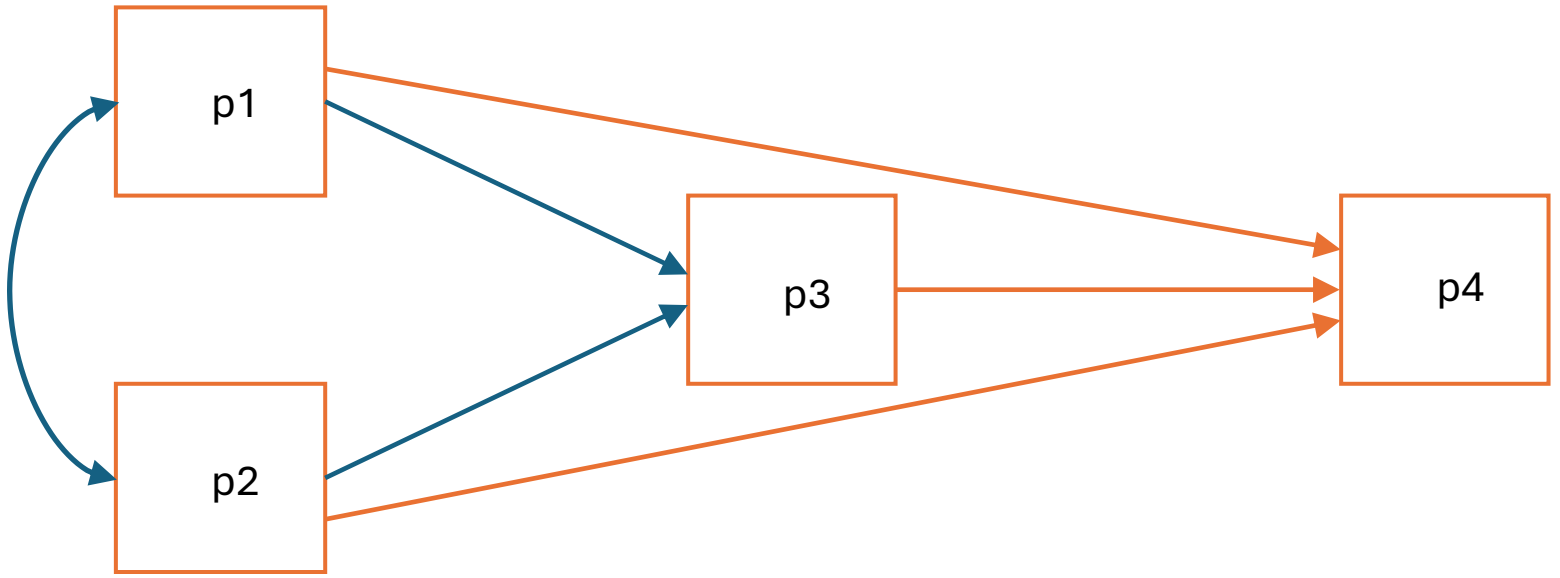


Syntax in *R* (*lavaan*)

p4 ~ p1 + p2 + p3

p3 ~ p1 + p2

p1 ~~ p2

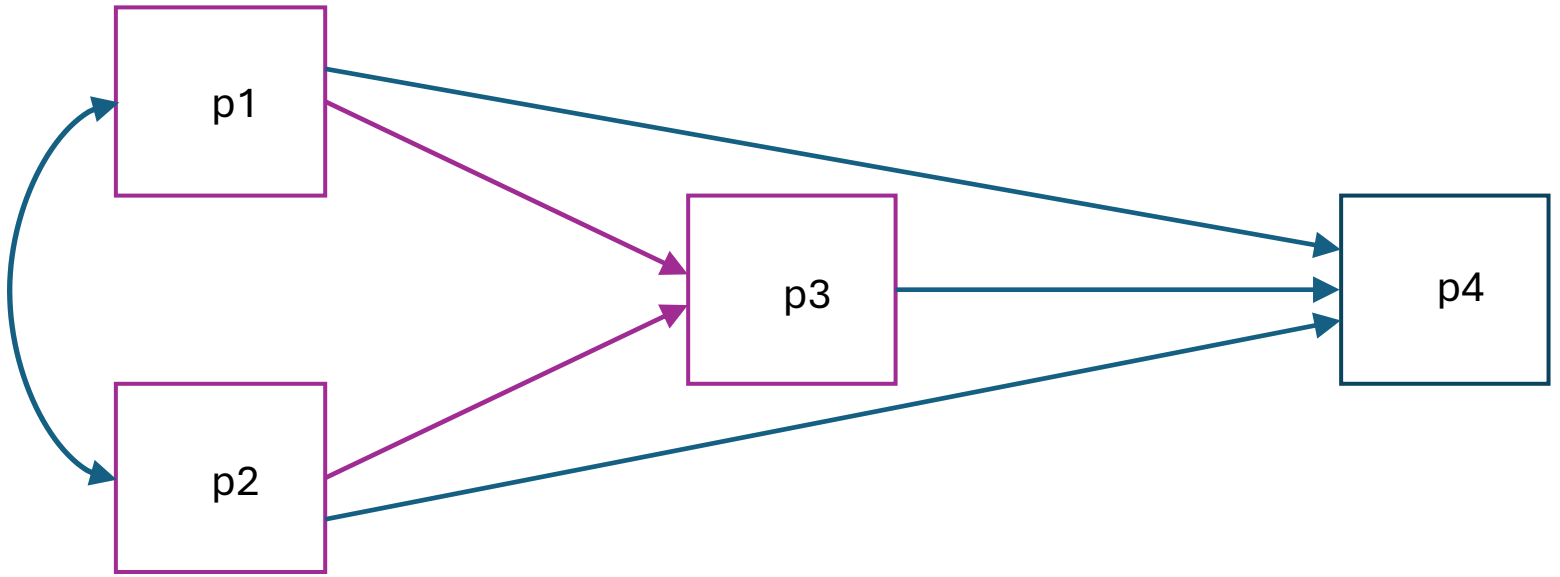


Syntax in *R* (*lavaan*)

p4 ~ p1 + p2 + p3

p3 ~ p1 + p2

p1 ~~ p2

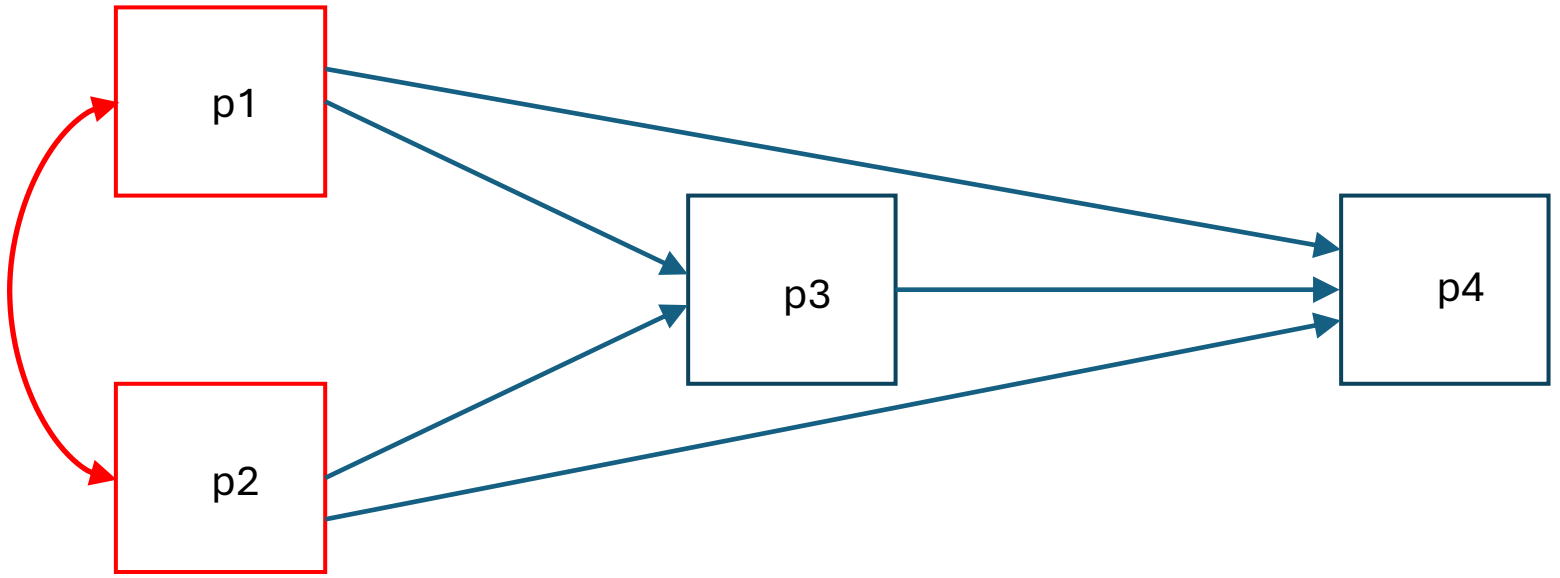


Syntax in *R* (*lavaan*)

p4 ~ p1 + p2 + p3

p3 ~ p1 + p2

p1 ~~ p2



Syntax in *R* (*lavaan*)

p4 ~ p1 + p2 + p3

p3 ~ p1 + p2

p1 ~~ p2

IT'S JUST LIKE HIERARCHICAL MULTIPLE REGRESSION

- Exogenous variable: variance assumed to be caused entirely by variables not in the causal model.

- Endogenous variable: variance assumed to be explained in part by other variables in the model.

- Each endogenous variable has a MR predicting that variable (Y) from all other variables hypothesised to have direct effects on Y.
  - Not included in these MRs are any variables that indirectly affect Y (through one or more intervening variables)
- These beta weights from the MR, are the path coefficients shown in a typical path analysis figure.

LIMITATIONS OF PATH ANALYSIS

- Path analysis can evaluate causal hypotheses, but it cannot establish the direction of causality.
- Path analysis is of little use at the exploratory stage of research.
  - Useful when have clear hypotheses to test, or a small number of hypotheses to test within a single path diagram.
- Path analysis requires a causal progression across a path diagram. Cannot use it where there are feedback loops in hypotheses.
- Endogenous variables must be capable of being treated as on an interval scale. Nominal, ordinal measurement (e.g., dichotomous outcomes) require other special types of analyses.
  - Basically put, all of the relationships in the path diagram must be capable of being tested by straightforward multiple regression. The intervening variables all have to serve as dependent variables in multiple regression analyses.

Model-based testing: Structural Equation Modeling

Part 3 of 4

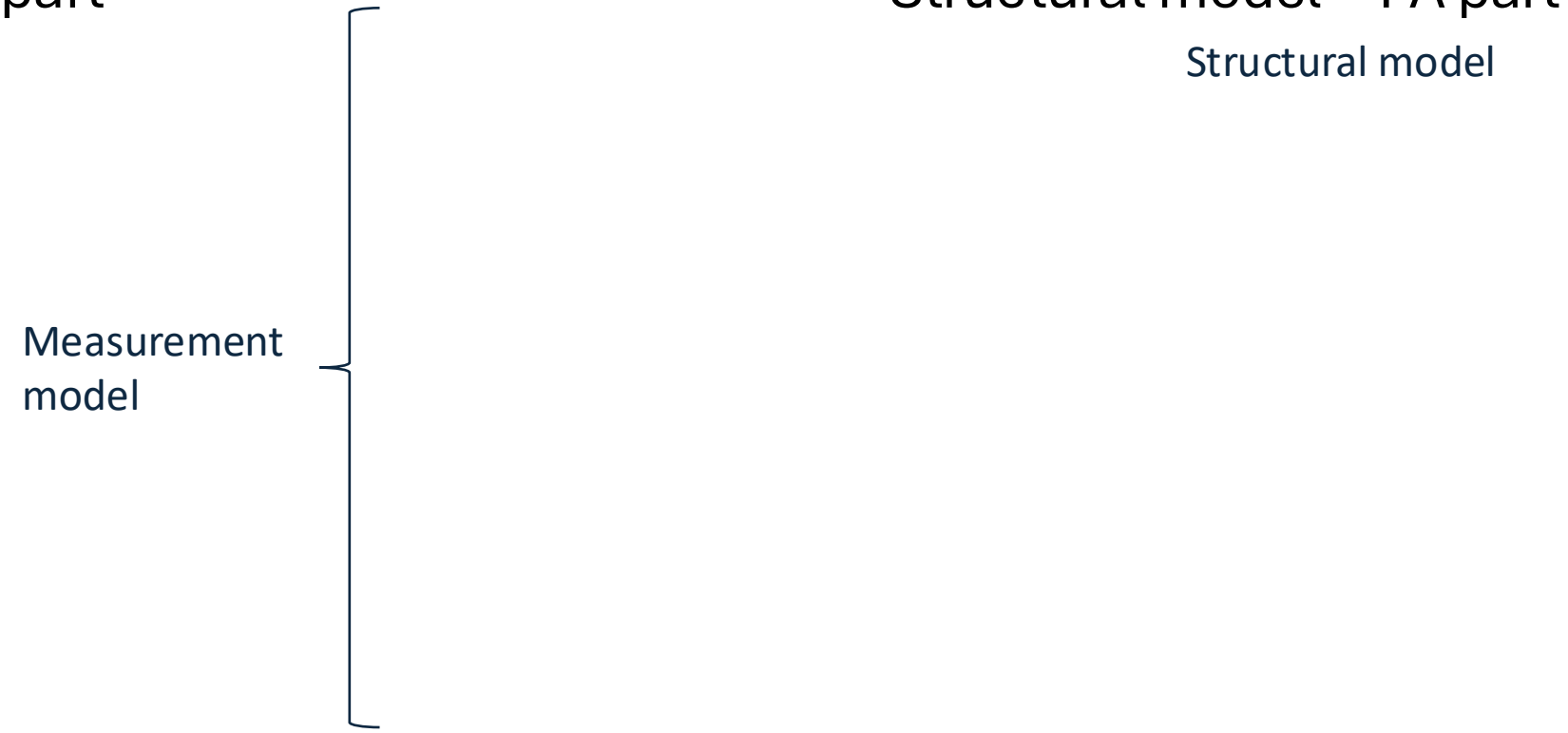
STRUCTURAL EQUATION MODELING

- Statistical framework that involves specification of a theoretical model, potentially involving latent/unobserved variables, and testing of this model with observed data.
  - e.g., CFA, Path Analysis, Latent Growth Modeling, RI-CLPM.
- Has been around for over 60years, but grown in popularity due to advances in computing.
- Model specification -> Estimation of free parameters -> Assessment of model and model fit (Model modification) -> Interpretation

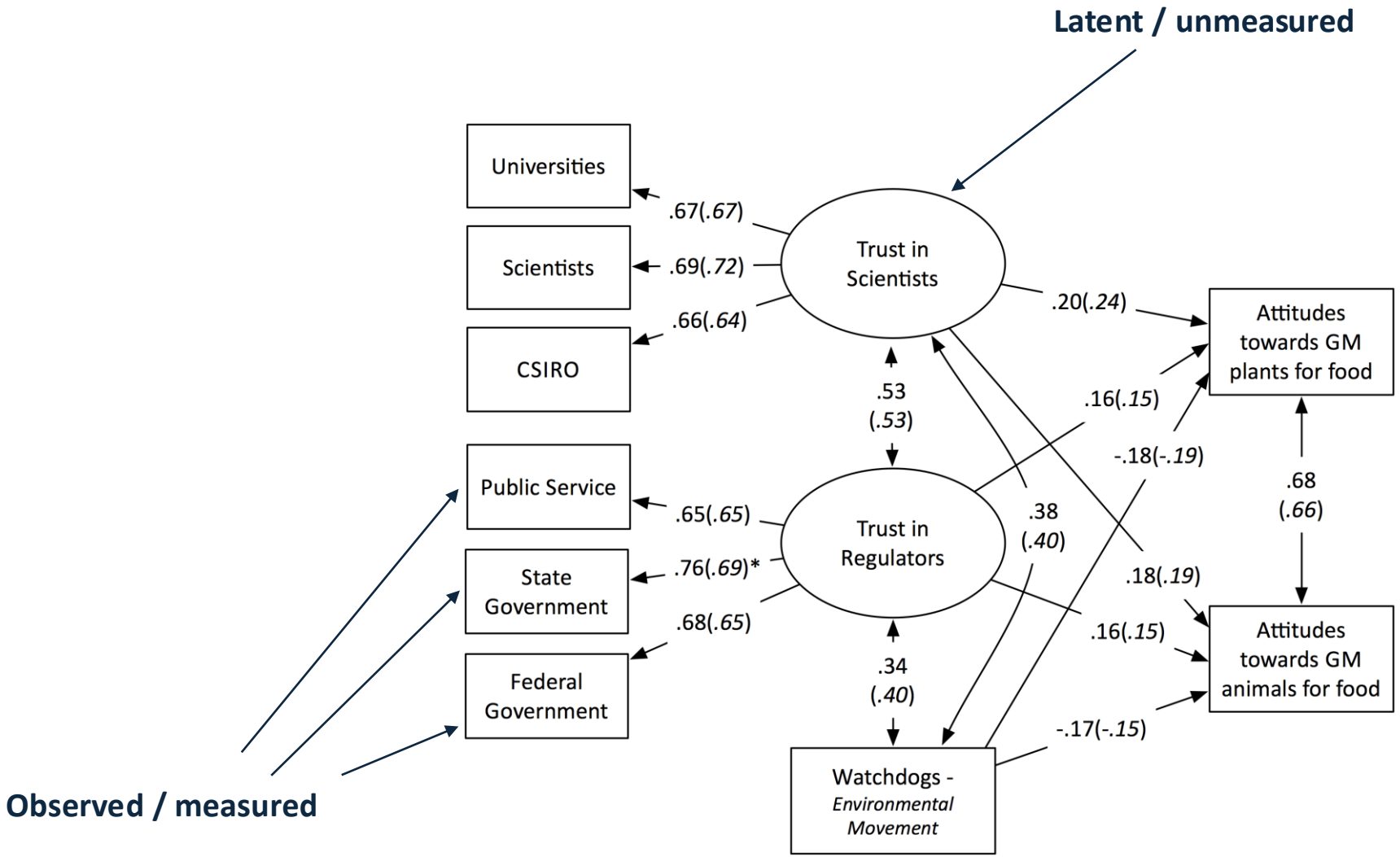
see Bauldry, S. (2015). Structural equation modeling. *International Encyclopedia of the Social & Behavioral Sciences* (2nd Ed.). Wright JD. Oxford: Elsevier, 615-620.

STRUCTURAL EQUATION MODEL

- A combination of CFA and PA
- Measurement model = CFA part
- Structural model = PA part



STRUCTURAL EQUATION MODEL



Marques, M. D., Critchley, C. R., & Walshe, J. (2015). Attitudes to genetically modified food over time: How trust in organizations and the media cycle predict support. *Public Understanding of Science*, *24*(5), 601-618.

THE FIT OF THE MODEL IN CFA

- The overall fit of the model is evaluated by using several fit indices (to compensate for limitations of each index):

Index	Recommended cut-off	
χ^2	Nonsignificant	
χ^2/ df	< 5.0 [a]	
RMSEA	< .08 [b, c]	
SRMR	< .08 [b, c]	
CFI	> .95 [b]	> .93 [d]
TLI	> .95 [b]	> .90 [d]

[a] Wheaton, Muthén, Alwin & Summers (1977)
[b] Putnick & Bornstein (2016)
[C] Hu & Bentler (1999)
[d] Byrne (1994)

Model-based testing: Putting it into action

Part 4 of 4

CONFIRMATORY FACTOR ANALYSIS

- Let’s practise conducting a CFA on a known series of constructs representing personality from the BIG-5 model with the dataset “bfi_big.omv”.

- We will model a 5-factor model, with each factor (latent variable) indicated by 5-items.

- Then we will evaluate the overall fit of the model.

BIG FIVE INVENTORY (BFI) - PERSONALITY

Agreeableness	Conscientiousness	Extraversion	Neuroticism	Openness
Am indifferent to the feelings of others. (R)	Am exacting in my work.	Don't talk a lot. (R)	Get angry easily.	Am full of ideas.
Inquire about others" well-being.	Continue until everything is perfect.	Find it difficult to approach others. (R)	Get irritated easily.	Avoid difficult reading material. (R)
Know how to comfort others.	Do things according to a plan.	Know how to captivate people.	Have frequent mood swings.	Carry the conversation to a higher level.
Love children.	Do things in a half-way manner. (R)	Make friends easily.	Often feel blue.	Spend time reflecting on things.
Make people feel at ease.	Waste my time. (R)	Take charge.	Panic easily.	Will not probe deeply into a subject. (R)

The items are from the IPIP (Goldberg, 1999). The data are from the SAPA project (Revelle, Wilt, & Rosenthal, 2010) , collected Spring, 2010

HISTORY: WHO WAS THE TALL POPPY ?

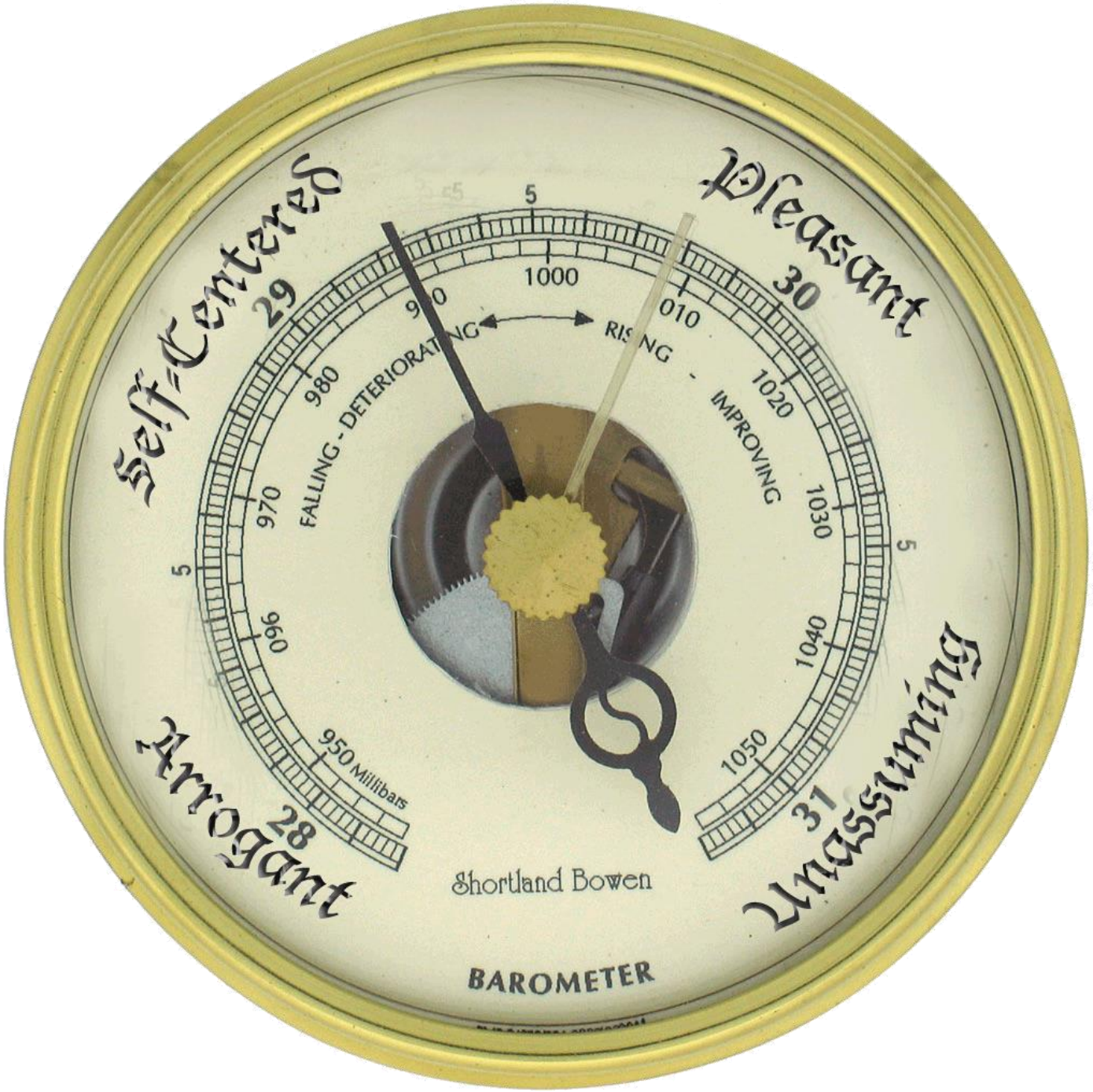
- Roman historian Livy (60BC-17AD) recounted story of the Roman ruler Tarquinius who walked around his garden cutting tall poppies with his stick.
- Distinctive roots within Anglo–Celtic origins of Australia.
- Nielsen (1902): tall poppies were people who were desired to retrench – burden on payroll (high income earner).
- Hancock (1930): Australian society provided "a 'fair and reasonable' standard of living for everyone, accompanied by a distrust for special excellence and a dislike of authority and status seekers.
- Term has shifted from being a reference to a high-income earner to a public figure of status.



https://commons.wikimedia.org/wiki/File:Poppies_Field_in_Flanders.jpg

WHAT IS A TALL POPPY ?

- OED: an especially well-paid, privileged, or distinguished person
- AND: person who is conspicuously successful" and frequently as "one whose distinction, rank, or wealth attracts envious notice or hostility"
- Tall Poppy Syndrome
- Arrogance Barometer



TALL POPPIES

Favour Fall	Favour Reward
2. It's good to see very successful people fail occasionally.	1. People who are very successful deserve all the rewards they get for their achievements.
3. Very successful people often get too big for their boots.	4. People who are very successful in what they do are usually friendly and helpful to others.
5. At school it's probably better for students to be near the middle of the class than the very top student.	6. People shouldn't criticise or knock the very successful.
7. Very successful people who fall from the top usually deserve their fall from grace.	9. The very successful person should receive public recognition for his/her accomplishments.
8. Those who are very successful ought to come down off their pedestals and be like other people.	11. One should always respect the person at the top.
10. People who are "tall poppies" should be cut down to size.	12. One ought to be sympathetic to very successful people when they experience failure and fall.
13. Very successful people sometimes need to be brought back a peg or two, even if they have done nothing wrong.	14. Society needs a lot of very high achievers.
15. People who always do a lot better than others need to learn what it's like to fail.	16. People who are right at the top usually deserve their high position.
19. Very successful people usually succeed at the expense of other people.	17. It's very important for society to support and encourage people who are very successful.
20. Very successful people who are at the top of their field are usually fun to be with.	18. People who are very successful get too full of their own importance.

Feather (1989)

NEED FOR INTELLECTION

- Intellection is a construct denoting individual differences in the degree to which one experiences either positive or negative emotional arousal while engaged in specific epistemic activities such as conceptual integration (Marques et al., 2017).
- ‘Intellection’ is defined as the act or process of using the intellect; thinking or reasoning that results in affective arousal.
- Intellection is also argued to be distinct from intelligence, and unlike need for cognition should be considered independent of intelligence (Cacioppo & Petty, 1982).
- This is consistent with Hofstadter's (1963) distinction between the Mental Technician and the Intellectual who may be both highly intelligent, but it is the latter who lives for ideas whereas for the former, ideas function as extrinsic means to other practical ends.

NEED FOR INTELLECTION

Working on difficult intellectual problems is enjoyable and stimulating for me.
I generally find physical or recreational activities more satisfying than intellectual activities. (R)
I tend to feel somewhat bored and impatient when dealing with remote, theoretical problems. (R)
Intellectual discovery is ok, but I prefer other forms of excitement. (R)
I'm probably the sort of person who would find it thrilling to be engrossed in a research project.
I deliberately seek out sources of intellectual stimulation
I have more exciting things to do than sit around and think all day long. (R)
I feel compelled to work on conceptual problems, even when I don't have to.
One of my favourite activities is discovering alternative ways of explaining a particular phenomenon.
The process of examining a concept in great detail is generally unappealing to me. (R)

Marques et al. (2017)

STRUCTURAL EQUATION MODEL

- Let's practise conducting an SEM using the dataset "NFI_TPS.omv".

- We will model a simple prediction that:

  Need for Intellection (NFI) → Favour Fall Tall Poppies.

- Then we will evaluate the overall fit of the model and the interpretation of the paths.

REFERENCES

Barrett, P. (2007). Structural equation modelling: Adjudging model fit. *Personality and Individual differences, 42*(5), 815-824. https://doi.org/10.1016/j.paid.2006.09.018

Bauldry, S. (2015). Structural equation modeling. *International Encyclopedia of the Social & Behavioral Sciences* (2nd Ed.). Wright JD. Oxford: Elsevier, 615-620.

Bollen, K. A. (1989). *Structural equations with latent variables*. Wiley.

Byrne, B. M. (2006). *Structural equation modeling with EQS: Basic concepts, applications, and programming* (2nd ed (Online-ausg.)). Lawrence Erlbaum Associates.

Cacioppo, J. T., & Petty, R. E. (1982). The need for cognition. *Journal of Personality and Social Psychology*, *42*(1), 116. https://doi.org/10.1037/0022-3514.42.1.116

Fan, X., Thompson, B., & Wang, L. (1999). Effects of sample size, estimation methods, and model specification on structural equation modeling fit indexes. *Structural Equation Modeling: A Multidisciplinary Journal*, *6*(1), 56–83. https://doi.org/10.1080/10705519909540119

Feather, N. T. (1989). Attitudes towards the high achiever: The fall of the tall poppy. *Australian Journal of Psychology*, *41*(3), 239–267. https://doi.org/10.1080/00049538908260088

Hofstadter, R., Wilentz, S., Hofstadter, R., & Hofstadter, R. (2020). *Anti-intellectualism in American life: The paranoid style in American politics: Uncollected essays, 1956-1965*. The Library of America.

Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: conventional criteria versus new alternatives. Structural Equation Modeling, *6*(1), 1–55. https://doi.org/10.1080/10705519909540118

Marques, M. D., Critchley, C. R., & Walshe, J. (2015). Attitudes to genetically modified food over time: How trust in organizations and the media cycle predict support. *Public Understanding of Science,* 24(5), 601–618. https://doi.org/10.1177/0963662514542372

Marques, M. D., Elphinstone, B., Critchley, C. R., & Eigenberger, M. E. (2017). A brief scale for measuring Anti-Intellectualism. *Personality and Individual Differences*, *114*, 167–174. https://doi.org/10.1016/j.paid.2017.04.001

Putnick, D. L., & Bornstein, M. H. (2016). Measurement invariance conventions and reporting: The state of the art and future directions for psychological research. *Developmental Review*, *41*, 71–90. https://doi.org/10.1016/j.dr.2016.06.004

Revelle, W., Wilt, J., & Rosenthal, A. (2010). Individual differences in cognition: New methods for examining the personality-cognition link. In A. Gruszka, G. Matthews, & B. Szymura (Eds.), *Handbook of Individual Differences in Cognition: Attention, Memory, and Executive Control* (pp. 27–49). Springer. https://doi.org/10.1007/978-1-4419-1210-7_2

Tabachnick, B. G., & Fidell, L. S. (2014). *Using multivariate statistics* (Pearson new international edition, sixth edition). Pearson.

Wheaton, B., Muthen, B., Alwin, D. F., & Summers, G. F. (1977). Assessing Reliability and Stability in Panel Models. *Sociological Methodology*, *8*, 84. https://doi.org/10.2307/270754